

MATH70102 UNSUPERVISED LEARNING

LIVE SESSION — WEEK 5

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IMPERIAL

Review of this week's content

Computational examples

Problem sheet questions

Density estimation

Overview

- Analyse the **distribution** of a data set, assuming that it can be described by a **probability density function** (pdf)
- **Parametric estimation**: seek a pdf from a **parametric family of pdfs**,

$$g(\cdot; \theta), \quad \theta \in \Theta$$

- **Non-parametric estimation**: seek a pdf from the **universe of all pdfs** (in principle)
- Should not be thought of as mutually exclusive — often useful to employ both approaches

Maximum likelihood estimation

Setup

- Suppose that the data $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$ are generated by **independent and identically distributed** d -dimensional random vectors $\mathbf{X}_1, \dots, \mathbf{X}_n$ with pdf $g_{\mathbf{X}}$
- Consider a parametric family of pdfs $g(\cdot; \boldsymbol{\theta})$, $\boldsymbol{\theta} \in \Theta$, such that

$$g_{\mathbf{X}}(\cdot) = g(\cdot; \boldsymbol{\theta}_0)$$

for some **unique** $\boldsymbol{\theta}_0 \in \Theta$

- The **joint pdf** of $\mathbf{X}_1, \dots, \mathbf{X}_n$ is

$$g_{\mathbf{X}_1, \dots, \mathbf{X}_n}(\mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_{i=1}^n g_{\mathbf{X}}(\mathbf{x}_i) = \prod_{i=1}^n g(\mathbf{x}_i; \boldsymbol{\theta}_0)$$

Maximum likelihood estimation (cont'd)

Likelihood function and maximum likelihood estimate

- We obtain the **likelihood function** from the joint pdf by **fixing** $\mathbf{x}_1, \dots, \mathbf{x}_n$ and **releasing** $\boldsymbol{\theta}_0$:

$$L(\boldsymbol{\theta}) = L(\boldsymbol{\theta}, \mathbf{x}_1, \dots, \mathbf{x}_n) = \prod_{i=1}^n g(\mathbf{x}_i; \boldsymbol{\theta}), \quad \boldsymbol{\theta} \in \Theta$$

- **Maximum likelihood estimate** (MLE) maximises the likelihood function:

$$\hat{\boldsymbol{\theta}} := \arg \max_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta}).$$

Maximum likelihood estimation (cont'd)

Log-likelihood function

- Usually more convenient to work with the **log-likelihood function**:

$$\ell(\boldsymbol{\theta}) := \log L(\boldsymbol{\theta}) = \sum_{i=1}^n \log g(\mathbf{x}_i; \boldsymbol{\theta}), \quad \boldsymbol{\theta} \in \Theta,$$

- Since $x \mapsto \log x$ is **strictly increasing**, log-likelihood is **equivalent** to likelihood for the purposes of optimisation:

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta}) = \arg \max_{\boldsymbol{\theta} \in \Theta} \ell(\boldsymbol{\theta}).$$

Maximum likelihood estimation (cont'd)

Uniqueness

- MLE **need not be unique** in general
- But lack of uniqueness may indicate a **problem with the model** (parameterisation) **or data**

Terminology: estimate vs. estimator

- **Estimate**: $\hat{\theta} = \hat{\theta}(\mathbf{x}_1, \dots, \mathbf{x}_n)$ (fixed value)
- **Estimator**: $\hat{\theta} = \hat{\theta}(\mathbf{X}_1, \dots, \mathbf{X}_n)$ (random variable/vector)

Maximum likelihood estimation (cont'd)

Finding MLE

- **Analytically:** by **calculus** (sometimes possible, but not always)
- **Numerically:** by **optimisation algorithms** (e.g., Newton's method or some more sophisticated variant such as L-BFGS-B)

Maximum likelihood estimation (cont'd)

Example (Multivariate normal distribution)

- Let $g_{\mathbf{x}}(\cdot; \boldsymbol{\theta}) = \phi(\cdot; \boldsymbol{\mu}, \Sigma)$ be the pdf of the **multivariate normal** with **mean vector** $\boldsymbol{\mu}$ and **covariance matrix** Σ
- Assume Σ is **positive definite** (**invertible**)
- Then:

$$\ell(\boldsymbol{\mu}, \Sigma) = -\frac{n}{2} \log(\det(\Sigma)) - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}_i - \boldsymbol{\mu})$$

- Deleted term $-\frac{nd}{2} \log(2\pi)$, which **does not affect maximisation**

Maximum likelihood estimation (cont'd)

Example (Multivariate normal distribution, cont'd)

- Maximising the log-likelihood $\ell(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ **analytically**, we find:

$$\hat{\boldsymbol{\mu}} = \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \quad (\text{sample mean})$$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \quad (\text{sample cov. matrix})$$

Maximum likelihood estimation (cont'd)

Asymptotic theory

Under **general conditions** one can show:

- **Consistency**: MLE $\hat{\theta}$ converges to the true value θ_0 as $n \rightarrow \infty$
- **Asymptotic normality**: Rescaled estimation error

$$\sqrt{n}(\hat{\theta} - \theta_0)$$

is **approximately normally distributed** when n is large
(useful for **inference**)

Estimation of mixtures

Overview

- For some **complex data sets**, standard families of pdf may **not** be flexible enough to fit the data
- Data sets with **multiple clusters** (necessitating a **multimodal** distribution) are an example
- We can construct more flexible distributions from **mixtures** of standard pdfs (e.g., normal)
- Estimated using the **EM algorithm**
- Bridge a gap between parametric estimation of (standard) pdfs and **non-parametric estimation**

Estimation of mixtures (cont'd)

Mixture density

- For parametric **component** pdfs $g_1(\cdot; \theta_1), \dots, g_q(\cdot; \theta_q)$ (where $q \geq 2$), define a **mixture pdf**

$$h(\mathbf{x}) := h(\mathbf{x}; \theta_1, \dots, \theta_q, \pi_1, \dots, \pi_q) := \sum_{j=1}^q \pi_j g_j(\mathbf{x}; \theta_j), \quad \mathbf{x} \in \mathbb{R}^d,$$

with **weights** $\pi_1, \dots, \pi_q > 0$ such that $\sum_{j=1}^q \pi_j = 1$

- Components often chosen from **same family**:

$$g_j(\cdot; \theta_j) = g(\cdot; \theta_j), \quad j = 1, \dots, q,$$

Estimation of mixtures (cont'd)

Generative representation using latent variable

- Let $\mathbf{X}$ be a d -dimensional **random vector**
- Let Z be a **categorical (latent) random variable** in $\{1, \dots, q\}$
- Specify, via **conditional distributions**,

$$\mathbf{X} \mid \{Z = j\} \sim g_j(\cdot; \boldsymbol{\theta}_j), \quad j = 1, \dots, q,$$

and

$$\mathbb{P}[Z = j] = \pi_j, \quad j = 1, \dots, q$$

- Then $\mathbf{X}$ **follows the mixture pdf** h

Estimation of mixtures (cont'd)

EM algorithm

- Direct maximum likelihood estimation of mixtures is **troublesome**, but the **EM (Expectation–Maximisation) algorithm** provides a better way
- **Data**: $\mathbf{x}_1, \dots, \mathbf{x}_n$ generated by **independent** $\mathbf{X}_1, \dots, \mathbf{X}_n \sim h(\cdot)$
- **Latent variables**: $Z_1, \dots, Z_n$ with realised values $z_1, \dots, z_n$ (**not observable!**)
- Collect **all parameters**:

$$\boldsymbol{\theta} := (\underbrace{\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_q}_{\text{components}}, \underbrace{\pi_1, \dots, \pi_q}_{\text{weights}})$$

Estimation of mixtures (cont'd)

EM algorithm (cont'd)

- Joint likelihood:

$$L(\boldsymbol{\theta}; \mathbf{x}_1, \dots, \mathbf{x}_n, z_1, \dots, z_n) = \prod_{i=1}^n \prod_{j=1}^q (g_j(\mathbf{x}_i; \boldsymbol{\theta}_j) \pi_j)^{\mathbf{1}_{\{z_i=j\}}},$$

$$\text{where } \mathbf{1}_{\{z_i=j\}} = \begin{cases} 1, & z_i = j, \\ 0, & z_i \neq j \end{cases}$$

- Corresponding log-likelihood:

$$\ell(\boldsymbol{\theta}; \mathbf{x}_1, \dots, \mathbf{x}_n, z_1, \dots, z_n) = \sum_{i=1}^n \sum_{j=1}^q \mathbf{1}_{\{z_i=j\}} (\log g_j(\mathbf{x}_i; \boldsymbol{\theta}_j) + \log \pi_j)$$

Estimation of mixtures (cont'd)

EM algorithm (cont'd)

Iterate two steps (until convergence):

1. **Expectation**: With **previous** (or **initial** if $t = 0$) **parameter values** $\theta^{(t)}$, compute

$$\begin{aligned} Q(\theta, \theta^{(t)}) &:= \mathbb{E}_{\theta^{(t)}, \mathbf{x}_1, \dots, \mathbf{x}_n} [\ell(\theta; \mathbf{x}_1, \dots, \mathbf{x}_n, Z_1, \dots, Z_n)] \\ &= \sum_{i=1}^n \sum_{j=1}^q \gamma_{i,j}^{(t)} (\log g_j(\mathbf{x}_i; \theta_j) + \log \pi_j), \end{aligned}$$

where $\gamma_{i,j}^{(t)} := \mathbb{P}_{\theta^{(t)}, \mathbf{x}_1, \dots, \mathbf{x}_n} [Z_i = j]$, known as **responsibility** of component j for sample i

Estimation of mixtures (cont'd)

EM algorithm (cont'd)

2. **Maximisation**: Obtain **new parameter values** in $\theta^{(t+1)}$ by maximising $\theta \mapsto Q(\theta, \theta^{(t)})$:

$$\theta^{(t+1)} := \arg \max_{\theta} Q(\theta, \theta^{(t)})$$

Remark on convergence

- May converge to a mere **local maximiser**
- Thus, experimentation with **different initial values** is recommended

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Problem 1

Question

Derive the equation

$$\gamma_{ij}^{(t)} = \frac{\pi_j^{(t)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_j^{(t)}, \boldsymbol{\Sigma}_j^{(t)})}{\sum_{j'=1}^q \pi_{j'}^{(t)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_{j'}^{(t)}, \boldsymbol{\Sigma}_{j'}^{(t)})}, \quad i = 1, \dots, n, \quad j = 1, \dots, q.$$

in Example 2.20 in the Lecture Notes using [Bayes' theorem](#).

Problem 1

Solution

By definition, $\gamma_{i,j}^{(t)} = \mathbb{P}_{\boldsymbol{\theta}^{(t)}, \mathbf{x}_1, \dots, \mathbf{x}_n} [Z_i = j]$

Bayes' theorem says:

$$g_{\mathbf{u}|\mathbf{v}=\mathbf{v}}(\mathbf{u}) = \frac{g_{\mathbf{v}|\mathbf{u}=\mathbf{u}}(\mathbf{v})g_{\mathbf{u}}(\mathbf{u})}{\int g_{\mathbf{v}|\mathbf{u}=\mathbf{u}'}(\mathbf{v})g_{\mathbf{u}}(\mathbf{u}')d\mathbf{u}'}$$

Problem 1

Solution (cont'd)

Set:

$$\mathbf{U} := Z_i, \quad \mathbf{u} := j, \quad \mathbf{V} := (\mathbf{X}_1, \dots, \mathbf{X}_n), \quad \mathbf{v} := (\mathbf{x}_1, \dots, \mathbf{x}_n)$$

Then:

$$\underbrace{\mathbb{P}_{\boldsymbol{\theta}^{(t)}, \mathbf{x}_1, \dots, \mathbf{x}_n}[Z_i=j]}_{g_{\mathbf{U}|\mathbf{V}=\mathbf{v}}(\mathbf{u})} = \frac{\underbrace{\phi(\mathbf{x}_i; \boldsymbol{\mu}_j^{(t)}, \Sigma_j^{(t)})}_{g_{\mathbf{V}|\mathbf{U}=\mathbf{u}}(\mathbf{v})} \underbrace{\pi_j^{(t)}}_{g_{\mathbf{U}}(\mathbf{u})}}{\underbrace{\int_{\Sigma_{j'=1}^q} d\mathbf{u}'}_{\phi(\mathbf{x}_i; \boldsymbol{\mu}_{j'}^{(t)}, \Sigma_{j'}^{(t)})} \underbrace{g_{\mathbf{V}|\mathbf{U}=\mathbf{u}'}(\mathbf{v})}_{\phi(\mathbf{x}_i; \boldsymbol{\mu}_{j'}^{(t)}, \Sigma_{j'}^{(t)})} \underbrace{g_{\mathbf{U}}(\mathbf{u}')}_{\pi_{j'}^{(t)}}}$$

Problem 2

Question

- (a) Let $g(\cdot; \theta)$, $\theta \in \Theta$, be a parametric family of pdfs on $\mathbb{R}^d$, for which we have, given some data $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$, found a **unique maximum likelihood estimate** $\hat{\theta}$. Suppose that we reparameterise the family in a new parameter space Ψ using an invertible function $t : \Theta \rightarrow \Psi$ such that

$$\tilde{g}(\mathbf{x}; \psi) := g(\mathbf{x}; t^{-1}(\psi)), \quad \mathbf{x} \in \mathbb{R}^d, \quad \psi \in \Psi.$$

Show that $\hat{\psi} := t(\hat{\theta})$ is the **unique maximum likelihood estimate** of ψ .

Problem 2

Solution

Define the **log-likelihoods**

$$\ell(\boldsymbol{\theta}) := \ell(\boldsymbol{\theta}; \mathbf{x}_1, \dots, \mathbf{x}_n) := \sum_{i=1}^n \log g(\mathbf{x}_i; \boldsymbol{\theta})$$

$$\tilde{\ell}(\boldsymbol{\psi}) := \tilde{\ell}(\boldsymbol{\psi}; \mathbf{x}_1, \dots, \mathbf{x}_n) := \sum_{i=1}^n \log \tilde{g}(\mathbf{x}_i; \boldsymbol{\psi}) = \sum_{i=1}^n \log g(\mathbf{x}_i; t^{-1}(\boldsymbol{\psi}))$$

Note that

$$\tilde{\ell}(\boldsymbol{\psi}) = \ell(t^{-1}(\boldsymbol{\psi}))$$

Problem 2

Solution (cont'd)

Take $\psi \in \Psi$ such that $\psi \neq \hat{\psi} := t(\hat{\theta})$

Since t is **invertible**, it follows that

$$t^{-1}(\psi) \neq t^{-1}(t(\hat{\theta})) = \hat{\theta}$$

But $\hat{\theta}$ is the **unique maximiser** of ℓ , so

$$\tilde{\ell}(\psi) = \ell(t^{-1}(\psi)) < \ell(\hat{\theta})$$

Problem 2

Solution (cont'd)

Write $\widehat{\theta} = t^{-1}(t(\widehat{\theta}))$, so that

$$\tilde{\ell}(\psi) < \ell(\widehat{\theta}) = \ell(t^{-1}(t(\widehat{\theta}))) = \tilde{\ell}(t(\widehat{\theta})) = \tilde{\ell}(\widehat{\psi})$$

Hence, $\widehat{\psi}$ is the **unique** maximum likelihood estimate

Problem 2

Question

- (b) If we **reparameterise** the **multivariate normal distribution** using its mean vector μ and **precision matrix** Ω (i.e., the inverse of the covariance matrix), what are the maximum likelihood estimates of μ and Ω ? (Assume throughout that the covariance matrix is invertible.)

Problem 2

Solution (cont'd)

Set for any $\mathbf{x} \in \mathbb{R}^d$ and any invertible $A \in \mathbb{R}^{d \times d}$,

$$t(\mathbf{x}, A) := (\mathbf{x}, A^{-1})$$

The **precision matrix** Ω is related to the **covariance matrix** Σ via

$$\Omega = \Sigma^{-1}$$

Hence $(\boldsymbol{\mu}, \Omega) = (\boldsymbol{\mu}, \Sigma^{-1}) = t(\boldsymbol{\mu}, \Sigma)$

By (a), $(\widehat{\boldsymbol{\mu}}, \widehat{\Omega}) = t(\bar{\mathbf{x}}, \widehat{\Sigma}) = (\bar{\mathbf{x}}, \widehat{\Sigma}^{-1})$, where $\bar{\mathbf{x}}$ is the **sample mean** and $\widehat{\Sigma}$ is the **sample covariance matrix**, is the MLE

Problem 3

Question

- (a) Write an implementation of the **EM algorithm** for the **Gaussian mixture model** (Example 2.20 in the Lecture Notes) in the special case $d = 1$ and $q = 2$.
- (b) Draw 1000 independent samples from $N(0, 1)$ and 500 independent samples from $N(4, 1.5)$ and merge the samples into a single data set. Then **test your implementation** with this data set.

Problem 3

Solution

→ **Jupyter Notebook**